

p-QuantSandbox  
Learn quant finance by building and competing with your own strategies

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## Project Overview

Quantitative finance shapes how trillions of dollars move, yet for most students it stays locked behind heavy math and industry jargon. Across the 5Cs there is real curiosity about how financial models and quant systems work, especially among students at CMC and elsewhere who lack a technical entry point. QuantSandbox closes that gap.

QuantSandbox is an interactive web platform where students build a portfolio, choose a strategy (momentum, mean reversion, or factor-based), and backtest it against real historical market data. The app computes performance and risk metrics (returns, volatility, Sharpe ratio, maximum drawdown) and presents them through clear, interactive visualizations. A game mode ranks student strategies on a shared leaderboard by risk-adjusted return, turning abstract concepts into something learned by doing.

The core data analysis work involves cleaning historical price series, computing returns and rolling statistics, simulating each strategy's trades over time, and calculating standard risk-adjusted performance metrics. Strategies are rule-based models (e.g. moving-average crossovers for momentum, z-score thresholds for mean reversion). The goal is a polished, genuinely usable tool that makes quant finance intuitive for a cross-campus audience, technical or not.

## Tech Stack

- Languages: Python (backtesting and data), JavaScript/TypeScript (frontend).
- Data & analysis: pandas, NumPy for portfolio simulation, returns, and risk metrics.
- Backend: FastAPI or Flask to serve strategy results.
- Frontend: React with Recharts or Plotly for interactive charts.
- Market data: yfinance or Alpha Vantage API for historical prices.
- Collaboration: GitHub for version control, Notion for docs, Discord for communication.
- Stretch: An LLM API (e.g. Claude or OpenAI) for a simulated news feed.

## Dataset (if applicable)

The project uses publicly available historical market data (daily open/high/low/close/volume for equities and ETFs) pulled from yfinance (Yahoo Finance) or the Alpha Vantage API. This data is used to backtest strategies: prices drive simulated trades, and the resulting equity curves are used to compute performance and risk metrics. No personal or proprietary data is involved. If the stretch news feed is built, headlines are synthetically generated by an LLM and are clearly labeled as simulated.

## **Ethical Considerations**

Historical market data reflects survivorship bias (delisted companies often drop out of datasets) and past market regimes that may not repeat, so backtested results can overstate real-world performance; the app will state this clearly so students do not mistake a good backtest for a guaranteed strategy. The tool is strictly educational and is not investment advice, and we will label it as such to avoid encouraging risky real-money decisions. A possible dual-use concern is that a realistic simulator could be misread as a live trading signal, so we will keep it clearly framed as a sandbox for learning, not a recommendation engine.

## **Team Structure**

**Target Team Size: 5 students (including PM).**

**Subteam Structure: Two sub-teams working in parallel.**

- **Data & Strategy (Python):** backtesting engine, data pipeline, strategy logic - 2 members.
- **Frontend & Visualization (React):** portfolio builder UI, charts, leaderboard - 2 members.
- **Finance & Product:** strategy design, educational content, UX and testing - 1 member (open to beginners).

**Collaboration Frameworks: GitHub (code and ticketing), Notion (docs and planning), Discord (day-to-day communication).**

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## **Timeline**

List how you intend to break down the development of your project into timely chunks, and roughly how you intend to delegate tasks. This is just a rough sketch of the development timeline, and the specific tasks and deadlines may change as the project progresses

|                                                                                              |                                                      |
|----------------------------------------------------------------------------------------------|------------------------------------------------------|
| Week 1 (9/8): Kickoff, team norms, environment setup, scope agreement, split into sub-teams. | Before finals: Final demo and documentation/handoff. |
| Weeks 2-3: Build core backtesting engine (one strategy                                       |                                                      |

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| end to end) and data pipeline; frontend scaffolds the portfolio builder.                                                                 |  |
| Weeks 4-5: Connect backend and frontend into the first full loop (build portfolio, run backtest, see results); add remaining strategies. |  |
| Weeks 6-7: Performance and risk visualizations, in-app educational explanations, polish.                                                 |  |
| Weeks 8-9: Game mode and leaderboard; begin stretch goals if on track.                                                                   |  |
| Weeks 10-11: Testing, bug fixing, UX refinement, preparing final demo.                                                                   |  |

### **Project Member Requirements**

Requirements vary by sub-team so we can include members at different levels:

- Data & Strategy: Comfortable with Python; intro CS (e.g. CS5/CS70 or equivalent). Familiarity with pandas is a plus but can be learned on the project.
- Frontend: Basic JavaScript and willingness to learn React; some web experience helpful.
- Finance & Product: No programming prerequisite. Curiosity about finance and a willingness to learn quant concepts from scratch.

No prior finance background is required for any role. This project is designed to teach quant concepts as we build.